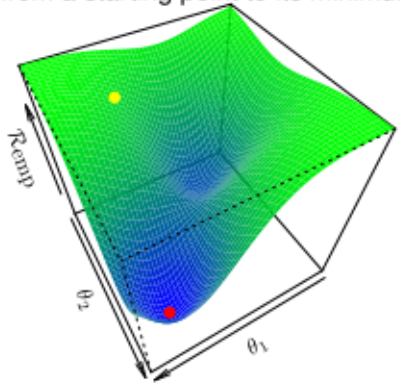
LEARNING AS PARAMETER OPTIMIZATION

- We can operationalize the search for a model f that matches training data best, by looking for its parametrization $\boldsymbol{\theta} \in \Theta$ with lowest empirical risk $\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$.
- Therefore, we usually traverse the error surface downwards; often by local search from a starting point to its minimum.

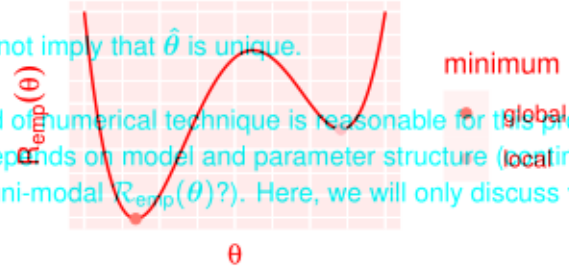


LOCAL MINIMA

If $\mathcal{R}_{\text{emp}}$ is continuous in θ we can define a **local minimum** $\hat{\theta}$:

$$\exists \epsilon > 0 \; \forall \theta \text{ with } \left\|\hat{\theta} - \theta\right\| < \epsilon \; \mathcal{R}_{\text{emp}}(\hat{\theta}) \leq \mathcal{R}_{\text{emp}}(\theta).$$

Clearly every global minimum is also a local minimum. Finding a local minimum is easier than finding a global minimum.



This does not imply that $\hat{\theta}$ is unique.

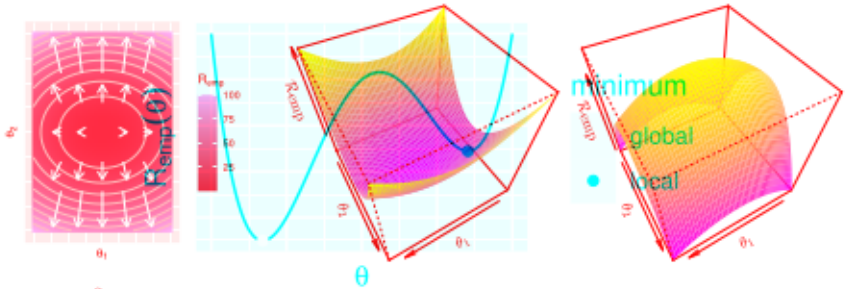
Which kind of numerical technique is reasonable for this problem strongly depends on model and parameter structure (continuous params? uni-modal $\mathcal{R}_{\text{emp}}(\theta)$?). Here, we will only discuss very simple scenarios.

LOCAL MINIMA AND STATIONARY POINTS

If $\mathcal{R}_{\text{emp}}$ is continuously differentiable in $\boldsymbol{\theta}$ then a **sufficient condition** for a local minimum is that $\hat{\boldsymbol{\theta}}$ is **stationary** with 0 gradient, so no local improvement is possible:

$$\exists \epsilon > 0 \; \forall \boldsymbol{\theta} \text{ with } \left\| \hat{\boldsymbol{\theta}} - \boldsymbol{\theta} \right\| < \epsilon : \frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\hat{\boldsymbol{\theta}}) = 0 \quad \mathcal{R}_{\text{emp}}(\hat{\boldsymbol{\theta}}) \leq \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}).$$

and the Hessian $\frac{\partial^2}{\partial \boldsymbol{\theta}^2} \mathcal{R}_{\text{emp}}(\hat{\boldsymbol{\theta}})$ is positive definite. While the neg. gradient points into the direction of fastest local decrease, the Hessian measures local curvature of $\mathcal{R}_{\text{emp}}$.



$\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$ — const. pos. def. Hessian — const. neg. def. Hessian

GRADIENT DESCENT: STATIONARY POINTS

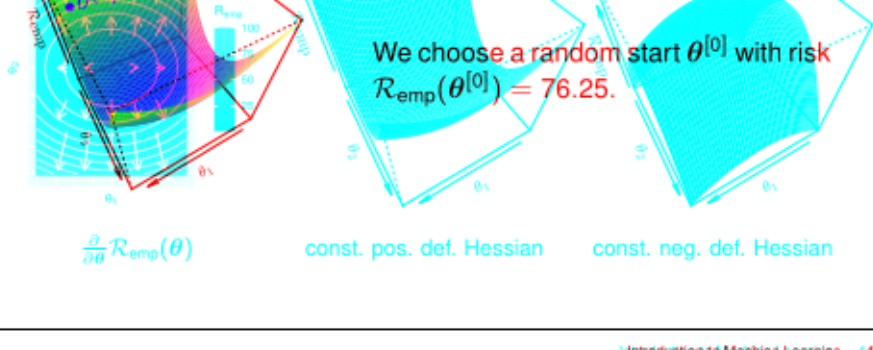
$\mathcal{R}_{\text{emp}}$ is a continuously differentiable in θ then a sufficient condition for a local minimum is that θ is **stationary** with 0 gradient, so no local improvement is possible:

$$\frac{\partial}{\partial \theta} \mathcal{R}_{\text{emp}}(\theta) = 0$$

and the Hessian $\frac{\partial^2}{\partial \theta^2} \mathcal{R}_{\text{emp}}(\theta)$ is positive definite. While the neg. gradient points into the direction of fastest local decrease, the Hessian measures local curvature of $\mathcal{R}_{\text{emp}}$.

The idea of GD is to iteratively go from the current candidate $\theta^{[t]}$ in the direction of the negative gradient, i.e., the direction of the steepest descent, with learning rate α to the next $\theta^{[t+1]}$:

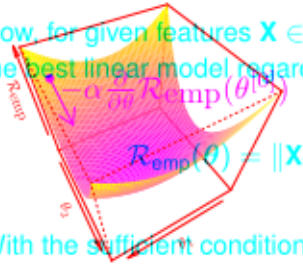
$$\theta^{[t+1]} = \theta^{[t]} - \alpha \frac{\partial}{\partial \theta} \mathcal{R}_{\text{emp}}(\theta^{[t]})$$



$\frac{\partial}{\partial \theta} \mathcal{R}_{\text{emp}}(\theta)$ const. pos. def. Hessian const. neg. def. Hessian

GRADIENT DESCENT – EXAMPLE

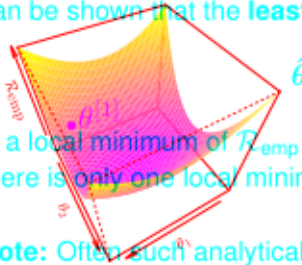
Now, for given features $\mathbf{X} \in \mathbb{R}^{n \times p}$ and target $\mathbf{y} \in \mathbb{R}^n$, we want to find the best linear model regarding the squared error loss, i.e.,



$$\mathcal{R}_{\text{emp}}(\theta) = \|\mathbf{X}\theta - \mathbf{y}\|_2^2 = \sum_{i=1}^{n} (\mathbf{x}^\top \theta - y^{(i)})^2 \ .$$

Now we follow in the direction of the negative gradient at $\theta^{[0]}$.

With the sufficient condition for continously differentiable functions it can be shown that the **least squares estimator**



$$\hat{\theta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

is a local minimum of $\mathcal{R}_{\text{emp}}$. If $\mathbf{X}$ is full-rank, $\mathcal{R}_{\text{emp}}$ is strictly convex and there is only one local minimum, which is also global.

We arrive at $\theta^{[1]}$ with risk $\mathcal{R}_{\text{emp}}(\theta^{[1]}) \approx 42.73$.
We improved: $\mathcal{R}_{\text{emp}}(\theta^{[1]}) < \mathcal{R}_{\text{emp}}(\theta^{[0]})$.

Note: Often such analytical solutions in ML are not possible, and we rather have to use iterative numerical optimization.

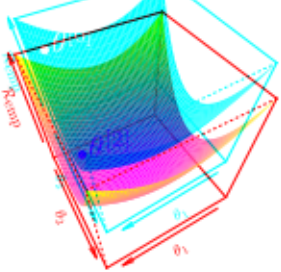
GRADIENT DESCENT - EXAMPLE

The simple idea of GD is to iteratively go from the current candidate $\theta^{[t]}$ in the direction of the negative gradient, i.e., the direction of the steepest descent, with learning rate α to the next $\theta^{[t+1]}$.

$$-\alpha \frac{\partial}{\partial \theta} \mathcal{R}_{emp}(\theta^{[1]})$$

Again we follow in the direction of the negative gradient, but now at $\theta^{[1]}$.

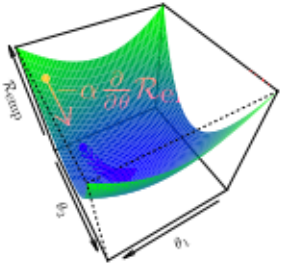
$$\theta^{[t+1]} = \theta^{[t]} - \alpha \frac{\partial}{\partial \theta} \mathcal{R}_{emp}(\theta^{[t]}).$$



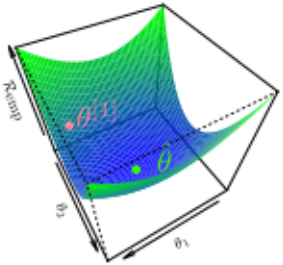
We choose a random start $\theta^{[0]}$ with risk $\mathcal{R}_{emp}(\theta^{[0]}) = 76.25$.

Now $\theta^{[2]}$ has risk $\mathcal{R}_{emp}(\theta^{[2]}) \approx 25.08$.

GRADIENT DESCENT - EXAMPLE



We iterate this until some form of convergence or termination.



We arrive at $\theta^{[1]}$ with risk

We arrive close to a stationary $\hat{\theta}$ which is hopefully at least a local minimum.

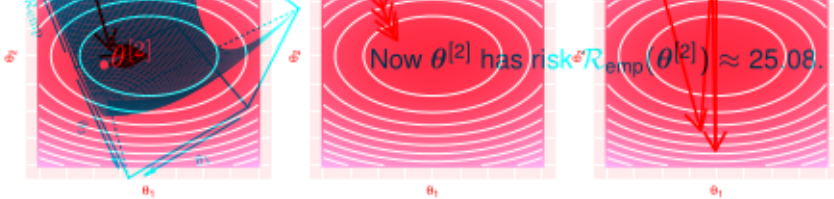
$\mathcal{R}_{\text{emp}}(\theta^{[1]}) < \mathcal{R}_{\text{emp}}(\theta^{[0]})$.

GRADIENT DESCENT - LEARNING RATE

- The negative gradient is a direction that looks locally promising to reduce $\mathcal{R}_{\text{emp}}$.
- Hence it weights components higher in which $\mathcal{R}_{\text{emp}}$ decreases more.
- However, the length of $-\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{R}_{\text{emp}}$ measures only the local decrease rate, i.e., there are no guarantees that we will not go "too far".
- We use a learning rate α to scale the step length in each iteration. Too much can lead to overstepping and no convergence, too low leads to slow convergence.
- Usually, a simple constant rate or rate-decrease mechanisms to enforce local convergence are used

$\boldsymbol{\theta}^{[2]} = \boldsymbol{\theta}^{[1]} - \alpha \frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[1]})$

Again we follow in the direction of the negative gradient, but now at $\boldsymbol{\theta}^{[1]}$.



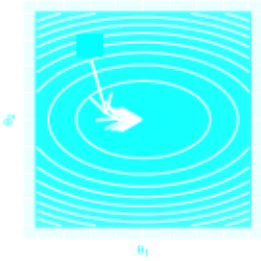
good convergence for α_1 | poor convergence for $\alpha_2 (< \alpha_1)$ | no convergence for $\alpha_3 (> \alpha_1)$

FURTHER TOPICS

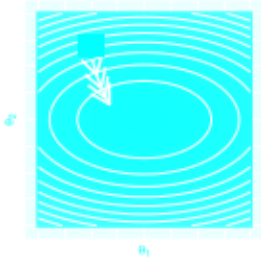
- Few models, e.g. linear regression, can be optimized analytically.
- GD is a so-called first-order method. Second-order methods (like Newton-Raphson) use the Hessian to refine the search direction for faster convergence.
- There exist many improvements of GD (momentum, ADAM), e.g., to smartly control the learn rate, to escape saddle points, to mimic second order behavior without computing the Hessian.
- If the gradient of is not computed on the complete data, but instead on small, random batches, this is **stochastic gradient descent** (SGD). For large-scale problems, this is usually more efficient.

GRADIENT DESCENT - LEARNING RATE

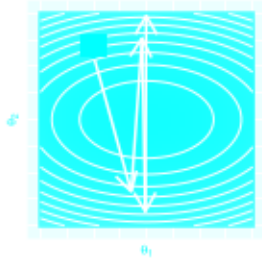
- The negative gradient is a direction that looks locally promising to reduce $\mathcal{R}_{\text{emp}}$.
- Hence it weights components higher in which $\mathcal{R}_{\text{emp}}$ decreases more.
- However, the length of $-\frac{\partial}{\partial\theta}\mathcal{R}_{\text{emp}}$ measures only the local decrease rate, i.e., there are no guarantees that we will not go "too far".
- We use a learning rate α to scale the step length in each iteration. Too much can lead to overstepping and no converge, too low leads to slow convergence.
- Usually, a simple constant rate or rate-decrease mechanisms to enforce local convergence are used



good convergence for α_1



poor convergence for $\alpha_2 (< \alpha_1)$



no convergence for $\alpha_3 (> \alpha_1)$

FURTHER TOPICS

- GD is a so-called first-order method. Second-order methods use the Hessian to refine the search direction for faster convergence.
- There exist many improvements of GD, e.g., to smartly control the learn rate, to escape saddle points, to mimic second order behavior without computing the expensive Hessian.
- If the gradient of GD is not derived from the empirical risk of the whole data set, but instead from a randomly selected subset, we call this **stochastic gradient descent** (SGD). For large-scale problems this can lead to higher computational efficiency.